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RAG + SLM/LLM Test Suite

Companion to *Foundations of Solar Microgrid Engineering*

This suite is designed to reveal **specific, diagnosable behaviors** — not just “did it answer correctly.” Every fact below was invented for this book, so if a model gets it right, it *must* have retrieved it from your RAG index rather than recalling it from pretraining. That’s the whole point: it lets you separate “the model is smart” from “the retrieval pipeline actually worked.”

Run the same questions across each model (llama3.2, phi4-mini, gemma3:4b, qwen3:8b, qwen3:14b) with and without RAG enabled, and log results in the scoring table at the end.

Category 1 — Direct Single-Chunk Retrieval (baseline sanity check)

Tests whether basic retrieval + answer extraction works at all.

#	Question	Expected Answer
1.1	What is the typical efficiency range of monocrystalline silicon panels?	20–23%
1.2	What percentage of a microgrid’s capital cost is typically battery storage?	30–35%
1.3	What does MPPT stand for?	Maximum Power Point Tracking
1.4	How many peak sun hours were assumed in the worked PV sizing example?	5 hours
1.5	What is the recommended inverter headroom multiplier over peak load?	1.25x (25% above peak load, minimum)

What to watch for: Small models often get this right if retrieval surfaces the correct chunk — this category mainly tests your *retriever*, not the *generator*.

Category 2 — Numerical / Formula Application

Tests whether the model can retrieve a formula AND correctly apply it to new numbers (not just parrot the example).

#	Question	Expected Answer / Reasoning
2.1	Using the book's formula, if a household needs 4,000 Wh/day and wants 3 autonomy days with a battery rated for 90% usable DOD, what battery capacity (Wh) is required?	$4,000 \times 3 \div 0.90 = \mathbf{13,333 \text{ Wh} (\sim 13.3 \text{ kWh})}$
2.2	Using the book's PV sizing formula, if daily demand is 1,800 Wh, peak sun hours are 4, and system efficiency is 80%, what PV capacity is required?	$1,800 \div (4 \times 0.80) = \mathbf{562.5 \text{ W}}$
2.3	A motor-driven load is rated at 500W running current. Per the book, what is the plausible surge range during motor start?	3–7x rated current (1,500W–3,500W)

What to watch for: This is where small models frequently fail even with correct retrieval — they retrieve the formula but botch the arithmetic. This is a strong SLM-vs-LLM differentiator.

Category 3 — Multi-Hop / Cross-Chapter Synthesis

Tests whether the model can combine facts that live in different chunks/chapters — the hardest thing for naive top-k RAG.

#	Question	Expected Answer
3.1	Which case study uses a battery chemistry chosen specifically for its non-flammability, and why was that important for that site?	Thornbury Cold-Storage Facility — sodium-ion battery, chosen due to food-storage safety requirements
3.2	Compare the battery chemistry used at Kestrel Ridge vs. Thornbury, and explain which one has the higher round-trip efficiency according to the chemistry comparison table.	Kestrel Ridge uses LFP (92–96% efficiency); Thornbury uses sodium-ion (80–85%). LFP is higher.
3.3	Which chapter's protection mechanism would be most relevant to the arc risks specifically mentioned in the ground-fault protection section?	Chapter 6.4, DC Arc-Fault Circuit Interrupters (AFCIs)
3.4	Of the three case studies, which has the largest PV-to-battery capacity ratio, and what does that suggest about its autonomy design relative to the others?	Meridian Village: 85kW/220kWh approx. 0.39; requires reasoning/calculation across the appendix table — good test of whether model does the division correctly

What to watch for: Small models tend to answer using only one retrieved chunk and ignore the second half of the question. This category is the best signal for whether you need a reranker or bigger top-k.

Category 4 — Negative/Out-of-Scope (hallucination check)

Tests whether the model will admit “not in the document” rather than inventing an answer.

#	Question	Expected Behavior
4.1	What is the maximum wind speed rating for the mounting structures described in this book?	Should say not covered / not specified in the document — this topic never appears in the book
4.2	According to the book, what brand of inverter is recommended?	No brand names are mentioned anywhere — model should say it's not specified
4.3	What is the current market price (2026) of an LFP battery per kWh?	The book gives <i>relative</i> cost only (“moderate”), never an absolute dollar figure — model should not invent a number
4.4	Does the book mention hydrogen fuel cells as a microgrid storage option?	No — should clearly say this is not discussed in the source material

What to watch for: This is the single most important test category. A model (especially larger ones with strong parametric knowledge) will often “helpfully” fill in a plausible-sounding but fabricated answer using outside knowledge instead of admitting the gap. Flag any response that gives a specific number/brand/spec not in the book as a **hallucination failure**, even if it sounds correct in the real world.

Category 5 — Ambiguous / Underspecified Questions

Tests whether the model asks for clarification vs. guessing, and how gracefully it handles ambiguity.

#	Question	What a good answer looks like
5.1	“What’s the best battery?”	Should surface that “best” depends on use case, and reference the comparison table dimensions (cost, cycle life, DOD) rather than picking one arbitrarily
5.2	“How big should my system be?”	Should ask for load data or explain that sizing depends on the 4-step process in Chapter 5, not give a single number
5.3	“Is this system safe?”	Should reference specific protection mechanisms (Ch. 6) rather than a blanket yes/no

Category 6 — Structured Data / Table Extraction

Tests whether the model can correctly read and reproduce tabular data — a common RAG weak point since tables often get mangled during chunking/parsing.

#	Question	Expected Answer
6.1	List all five battery chemistries in the comparison table, in the order they appear.	Lead-acid (flooded), Lead-acid (AGM/sealed), LFP, NCM, Saltwater (sodium-ion)
6.2	What is the cycle life range for NCM batteries specifically?	1,000–2,000 cycles

#	Question	Expected Answer
6.3	In the Appendix C case study table, which site uses “Hybrid, parallel inverters” as its architecture?	Meridian Village Cooperative

What to watch for: If your PDF parser/chunker mangles tables into unstructured text, models will frequently mix up rows/columns here even when the answer “should” be retrievable. Good diagnostic for your ingestion pipeline, not just the model.

Category 7 — Summarization / Compression

Tests output quality independent of retrieval — good for comparing writing quality across model sizes.

#	Prompt
7.1	Summarize Chapter 4 (Battery Chemistries) in 3 sentences.
7.2	Summarize the entire book in exactly 100 words.
7.3	Explain the difference between grid-forming and grid-following inverters to someone with no electrical background.

Scoring approach: These have no single “correct” answer — score on faithfulness (no invented facts), completeness, and whether small models drop nuance that larger ones keep.

Category 8 — Classification Task (for testing SLM-as-router/classifier use case)

Use these to test a small model’s ability to act as a lightweight classifier/router *before* invoking your larger LLM or RAG pipeline — the pattern discussed earlier (intent routing).

Prompt template:

Classify the following user query into exactly one category:

[factual_lookup, calculation_request, safety_question, out_of_scope, comparison_request]

Respond with only the category label.

Query: "{query}"

#	Query	Correct Label
8.1	“What is MPPT?”	factual_lookup
8.2	“If my daily load is 3kWh, how big a battery do I need for 2 days?”	calculation_request
8.3	“Is lithium battery storage a fire risk?”	safety_question
8.4	“What’s the weather like in Nairobi today?”	out_of_scope
8.5	“Which is better, LFP or sodium-ion, for a hot climate?”	comparison_request

What to watch for: This tests exactly the use case you asked about — run this across phi4-mini, gemma3:4b, and llama3.2 and measure: (a) accuracy against the label set, (b) whether the model outputs *only* the label or adds unwanted commentary (critical for programmatic parsing), and (c) latency, since this is meant to be a cheap pre-filter before your bigger model runs.

Suggested Scoring Table

Test #	llama3.2	phi4-mini	gemma3:4b	qwen3:8b	qwen3:14b	Notes
1.x avg						Basic retrieval
2.x avg						Arithmetic on retrieved formulas
3.x avg						Multi-hop synthesis
4.x (hallucination rate)						Lower is better — count fabricated answers
5.x (clarification quality)						Qualitative
6.x avg						Table extraction accuracy
7.x (writing quality)						Qualitative 1–5
8.x (classification accuracy)						% correct labels, format compliance

How to Interpret Results

- **If small models fail Category 1 but big models pass:** your retriever is fine, but small models struggle to extract/use retrieved context — consider better prompt templates or a rerank step.
- **If ALL models fail Category 3 (multi-hop):** the problem is likely your retriever’s top-k or chunk size, not the generator — try increasing top-k or adding a summary-index layer.
- **If big models fail Category 4 more than small ones:** this is common — larger models have more confident parametric knowledge and are more prone to “filling in” plausible answers instead of admitting a gap. Explicitly instruct in your system prompt: *“Only answer from the provided context. If the answer is not present, say so.”*
- **If small models pass Category 8 with clean single-label output:** that confirms they’re viable as a cheap classification/routing layer in front of your larger generation model, exactly the architecture discussed for your pipeline.